The Technology Value Stream: Lead Times

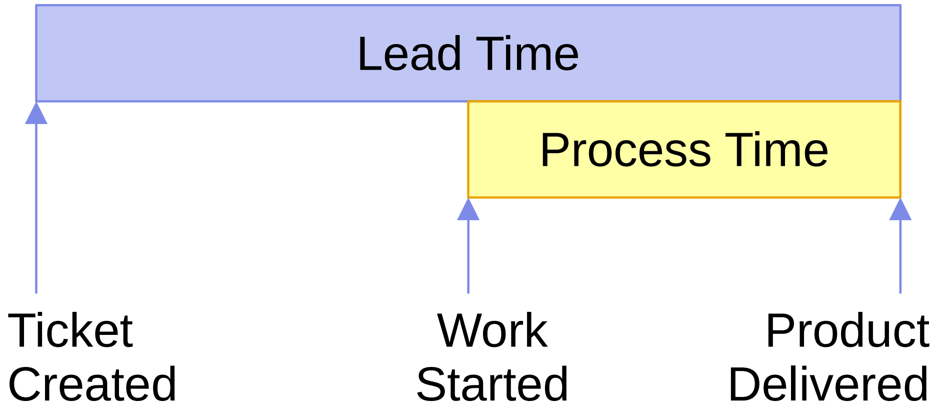
CSD380: DevOps
Module 1: The History of DevOps
Assignment 2: The Technology Value Stream

Lead Time Concepts

- Lead Time vs Process Time
- Traditionally: Deployment Lead Times Requiring Months
- Ideally: Deployment Lead Times Measured in Minutes

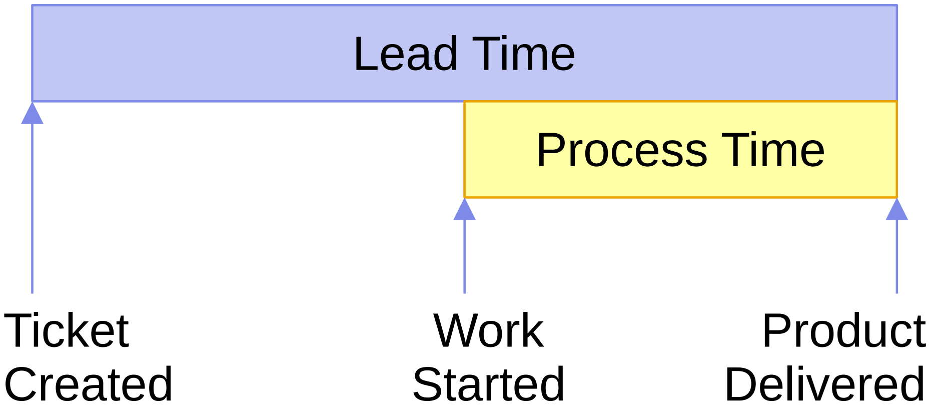
Lead Time vs Process Time

- Lead Time is the entire feature, from the moment the initial ticket is created
- Process Time is the time the actual development work takes – minus the time spent evaluating, estimating, and planning development.



Lead Time vs Process Time

- It is tempting to optimize Process Time – because you have an estimate that you can measure your outcome against
- But customers don't feel Process Time – It's not real to them!
- Optimizing the software process across the entire Lead Time makes sure the customer isn't kept waiting for no reason



Traditionally: Deployment Times Requiring Months

- Software complexity increases as the complexity of our needs increses
- We have developed wonderful tools and processes to deal with this complexity
- Now we need to develop strategies to deal with our tooling and process complexity!

Traditionally: Deployment Times Requiring Months

- Overly complex requirements can require lots of implementation time and detailed verification steps
- Manual or improvised verification and approval steps can put many roadblocks in front of verification
- Team specialization without effective communication can reduce efficiency and create integration problems
- Underspecified, unclear designs which lack hierarchy can prevent efficient allocation of resources, and create documentation and integration problems

Ideally: Deployment Times Measured in Minutes

- Keeping code changes and requirement scope small creates lean edits that can be easily understood, implemented, and verified
- Wise use of CI and CD systems to efficiently build, integrate, test, verify, and deploy a system can quickly catch issues that humans never would!
- Team specialization *with* effective collaboration still allows efficient team autonomy while seizing opportunities to use combined expertise
- Well-encapsulated designs, which have clean API boundaries, allow specialized teams to work efficiently, and lets modules be tested in isolation to contain errors

Sources

- admin. (2024, April 10). Lead time in DevOps: A key to efficient software delivery. Metridev. https://www.metridev.com/en/metrics/lead-time-in-devops-a-key-to-efficient-software-delivery/
- Kim, G., Humble, J., Debois, P., Willis, J., & Forsgren, N. (2021). The DevOps handbook, second edition. IT Revolution.

Thank You!